Who Rises in the Colony? Broken Promises, Memory and Public Records in a Society of LLM Agents[1]

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With
Apart Research

Abstract

Agent swarms are often built from copies of one model that differ only in a few lines of instructions. We ask which of those differences turn into influence, and what lets such a group hold a dishonest member to account. LLM agents share one system prompt for a fishing colony and differ only in five randomised trait sentences and two placebos. Across 234 sixteen-day games, 36 thousand-day games (3 colony set-ups per condition) and 40-day pilots on five models, we measure keeper votes, private trust polls and food, and plant a colonist who promises food every day and never gives it. Prompted traits reliably change behaviour, but we found no influence effect that replicates across models. What protects the cheat is what agents keep in memory: victims' self-written notebooks recorded the promise far more often than the missing food (26–71% against 2–7% on gemma-4-31b), and one added sentence stating that the food never arrived cuts their support by about two thirds. Over 1000 days, a public record of gifts shuts the promise-breaker out: it draws 0.3–1.5% of trust votes against 8.5–9.6% without the record, and every game with the record sits below every game without it. In a separate 40-day pilot, a private ledger achieved about half of the board's reduction in trust. Explicit, durable records of commitments and deliveries look like a cheap safeguard for multi-agent systems.

[1] Research conducted at the Swarm Alignment Hackathon, November 2026

1. Introduction

Large deployments of AI agents are increasingly populations of one model: the same weights, the same scaffold, and a few lines of role or persona text that differ from agent to agent. Safety evaluations mostly test those agents one at a time. In a population, however, influence is allocated socially: some agents get listened to, trusted and handed control of shared resources. If certain dispositions reliably rise, a swarm selects for them whether or not anyone intended it; and if a dishonest member can keep the group's trust, the group cannot police itself.

We study both questions in a small, fully logged society. Agents live in Larkspur, a fishing colony where food is the only currency, cooperation pays, a valuable skill can be hoarded or taught, and unannounced events ask the colony to choose a keeper for its winter store and, privately, whom it trusts. All agents share one system prompt and differ only in a few randomly assigned trait sentences. We then plant one colonist who publicly promises food to a hungry member every day and never gives it.

Our main contributions are:

1. **A paired testbed** for influence and accountability in LLM populations: seeded games are identical across conditions, so every intervention is compared game by game, and placebo traits calibrate chance findings.
2. **Traits shape behaviour, not reliably influence:** effects on behaviour replicate across two models; we found no effect on votes that does.
3. **Why deception goes unsanctioned:** agents' self-written notes keep the promise and lose the missing gift, and counterfactual replays show the same models withdraw support once non-delivery is stated.
4. **Over 1000 days, a record decides:** a public gift record shuts the promise-breaker out in all 9 paired long games (3 colony set-ups in 3 economies); in a separate 40-day pilot a private ledger achieves about half of that reduction.

2. Related Work

Populations of LLM agents form relationships and coordinate in open-ended worlds (Park et al. 2023), develop norms through dialogue (Horiguchi et al. 2024), and mostly fail to sustain a shared fishery (Piatti et al. 2024). Their cooperation depends on what they can observe: with a visible record of past donations, Claude 3.5 Sonnet used costly punishment to sustain cooperation (Vallinder & Hughes 2024); gossip and reputation systems restore cooperation that otherwise collapses (Zhu et al. 2026; Ren et al. 2025); and LLM agents may lack the grounding that reputation mechanisms assume (Hu et al. 2026). Whether LLMs punish is contested: they are often forgiving (Fontana et al. 2024; Akata et al. 2025) and prefer rewarding to punishing (Guzman Piedrahita et al. 2025), yet replicate human third-party punishment (Cross et al. 2025; Liu et al. 2025). LLMs break public promises for self-interest (Shi et al. 2026a) and can be exploited when they treat announcements as binding (Shi et al. 2026b); facing Byzantine agents in coordination games, they detect betrayal within a round but do not adapt (El Mir et al. 2026). In humans, indirect reciprocity runs on reputation (Nowak & Sigmund 2005; Wedekind & Milinski 2000).

Our position. Where LLM agents do sanction, the setting usually supplies explicit payoffs, a visible record of defection or an interaction history, and a dedicated punish action. In our colony a breach is visible only to its victim, each agent's memory is a short notebook it rewrites itself, sanctions must be improvised from votes, partnerships and gifts, and the defector keeps talking. Our focus is the interaction between unfulfilled commitments, lossy self-written memory and the allocation of trust, which we probe with replays and with records added one at a time (public, private, plus a punish action), over up to 1000 days.

3. Methods

The colony. Each agent is one of the N members of Larkspur and knows the rules of daily life, but nothing about scores, game length or upcoming events. Members start with 5 food and eat 2 a day. Fishing alone brings in 2 food, two members who name each other 3 each, an unreciprocated choice 1; weavers (a quarter at the start) add 2 for themselves and 1 for their partner, and can spend a day teaching the skill. Members can give food, post one public message and send two private messages a day. Each agent's only memory is an 80-word notebook it rewrites daily.

Traits. The 560-word system prompt is identical for all members except seven shuffled "About you" sentences: five traits (self-interest, assertive, guarded, reciprocity, loner; e.g. "You like working alongside others" versus "... on your own") and two placebos (birth season, favourite colour), each version given to a random half of every game. **Influence** is measured by unannounced events: a storm, a secret windfall, a vote for a keeper who divides the winter store (and keeps what it does not hand out), and a private poll of whom each member trusts most.

The promise-breaker. In treated games one member is privately told to publicly promise food to a hungry member every day, never give any, and if challenged blame a mix-up; the engine blocks its gifts. Its seat comes from a separate random stream, so each treated game is otherwise identical to its control game, and we compare the same seat across the two.

Interventions. The *gift board* shows yesterday's gifts to everyone; the *private ledger* shows each member only its own gifts given and received ("No one gave you food"); a *punish* action costs 1 food and removes 2 from the target. *Economies* are positive-sum (as above), zero-sum (a fixed catch divided by claimed shares) or negative-sum (the catch shrinks as more shares are claimed).

Designs. (i) 16-day games with colonies of 4, 8 and 16 on gemma-4-31b-it and gemma-4-26b-a4b-it: 234 games. (ii) 1000-day games crossing economy, gift board and promise-breaker: 3 games of 8 agents per condition on gemma-4-31b-it, each with 50 keeper terms and 50 trust polls (about 288,000 replies; spoilage above 15 food and a one-sentence reason per reply). (iii) 40-day pilots (3 colonies, a poll every 10 days, positive-sum, spoilage and reasons) on gemma-4-26b, gemma-4-31b, glm-5.2 and deepseek-v4-flash, a partial run on qwen3.5-397b (stopped at days 27–36), and private-ledger and punish arms on gemma-4-26b. All models via the Adaption API; every run uses seed 0.

Analysis. Trait effects are regression coefficients with game fixed effects and within-game permutation p-values; promise-breaker effects are paired differences with the same seat in control. *Counterfactual replays* re-ask a victim's logged vote with one change to its prompt (8 samples per variant; intervals resample whole games). *Notebook coding* classifies victims' next-day notebooks

by keyword. Criteria for four long-game "catches" (invented institution, persistent leader, sorting, reputation decline) and for the punish test were committed to git before the relevant runs were analysed.

4. Results

Traits change behaviour; which traits win influence depends on the model. On both models trait sentences shift behaviour ($p < 0.05$; e.g. loners do less paired work at every colony size), but no influence effect is significant on both, and the models agree on the direction of only 27 of 45; we did not test model-by-trait interactions, so this is an absence of a reproducible effect, not a demonstrated difference. Placebos passed 0 of 96 tests, fewer than the 5 expected by chance. Where traits pay, mediation suggests they pay through contact (partnerships and messages). Over 16 days the promise-breaker was not detectably sanctioned in any setting: a weak null, with one trust poll per game.

The models can sanction once told; the colony never records the breach. Adding one sentence to a victim's logged vote, "The food [name] promised you has never arrived", cuts support for the promise-breaker by about two thirds (Figure 1; gemma-4-31b 20.3% to 7.0%, —13.3 points, 95% interval resampling games [—22.8, —6.0]; gemma-4-26b 40.7% to 13.0%, [—37.9, —19.5], lower in all 18 games). This shows that stating non-delivery changes the decision, not by itself why the original decision went wrong: the sentence may also add salience or authority. The notebooks point to what is missing (Table 1): without a record, victims wrote down the promise far more often than the missing food, and on gemma-4-26b 60 of 117 notes claimed a gift that never happened ("Dov kept his promise and gave me food."), written by 5 of 6 victims, one of whom carried the claim forward on 36 days.

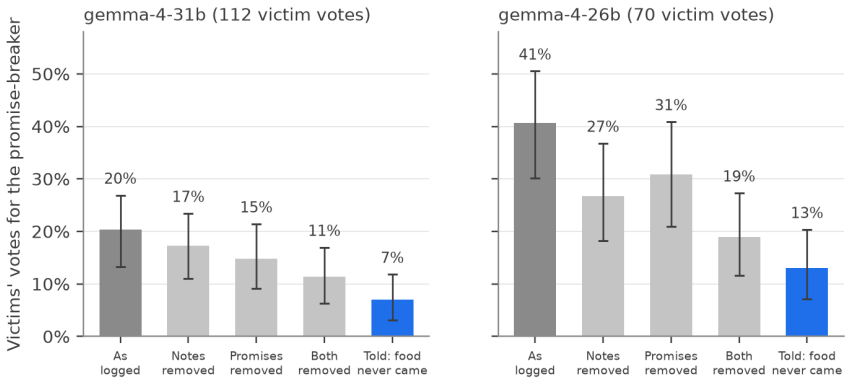


Figure 1. Counterfactual replays: each logged vote by a victim (112 on gemma-4-31b, 70 on gemma-4-26b) re-asked 8 times with one change to that day's prompt. Bars: share of votes for the promise-breaker, with 95% intervals. Stating that the food never arrived (blue) cuts support by about two thirds.

Table 1. Share of victims' next-day notebooks, after a public promise, that record the promise, the missing food, or a gift that never came (keyword coding; gemma-4-31b ranges span three economies), and the promise-breaker's share of trust votes (fair share 12.5%).

Setting	Notes record the promise	Notes record the missing food	Notes claim a gift	Trust share
gemma-4-31b, 1000 days, no record	26–71%	2–7%	about 1%	8.5–9.6%
gemma-4-31b, 1000 days, public board	43–52%	57–64%	under 1%	0.3–1.5%
gemma-4-26b, 40 days, no record	50%	2%	51%	20.8%
gemma-4-26b, 40 days, private ledger	62%	22%	26%	14.6%
gemma-4-26b, 40 days, public board	74%	14%	11%	7.3%

Over 1000 days, a public record decides. With the gift board the promise-breaker drew 0.3%, 0.8% and 1.5% of trust votes in the positive-, zero- and negative-sum economies, against 8.5%, 9.6% and 9.6% without it and about 8% for the same seat in control (Figure 2). Every game with the board (0.0–2.2%) sat below every game without it (4.5–13.8%), and it was keeper in none of 450 terms. The 9 pairs are 3 colony set-ups replayed in 3 economies; at the level of set-ups the effect is 3 of 3 (sign test $p = 0.25$), so the evidence is its size and consistency (the pair-level test, $p = 0.004$, was not pre-registered). The gap appears at the first poll in the zero- and negative-sum economies but takes about 100 days in the positive-sum one. Without a record only the zero-sum colony withdraws trust, slowly (13.8% in the first third to 6.2% in the last; slope —0.20 points a season, 95% interval [—0.29, —0.11]), the one pre-registered reputation catch.

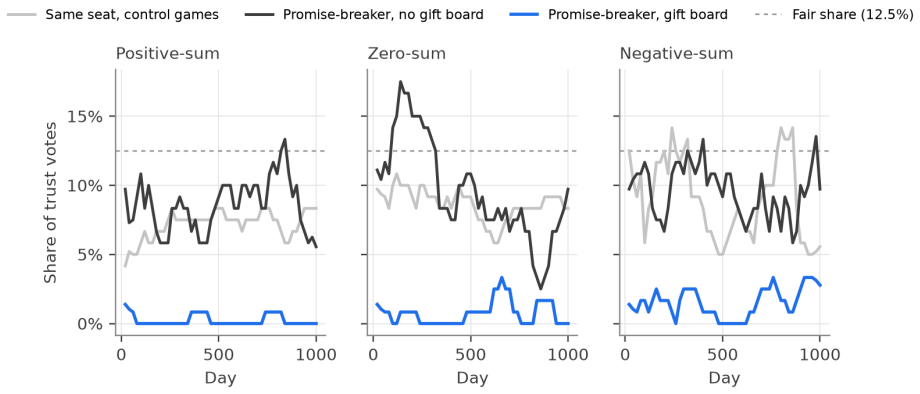


Figure 2. The promise-breaker's share of trust votes over 1000 days (gemma-4-31b, 8 agents, mean of 3 games, 5-poll moving average; per-game values in the text). With the gift board (blue) it is shut out in every economy; without it (black) it keeps about the standing of the same seat in control games (grey).

Publicity or memory, and other models. The board both publicises a breach and reminds its victim of it. With a private ledger instead (gemma-4-26b, 3 colonies, 40 days) the promise-breaker's trust share fell from 20.8% to 14.6%, against 7.3% with the board, in that order in every colony, and its keeper nominations fell from 16.7% to 5.2% with either record: the private ledger achieved about 46% of the board's reduction in trust. This is not a causal split between memory and publicity. The

treatments differ in more than one way, and the private ledger produced more notes of the missing food than the board (22% against 14%) while reducing trust less. At 40 days the board lowered its trust in every colony on gemma-4-26b and glm-5.2; deepseek-v4-flash sat near a 5% floor either way (its victims noted the missing food themselves), and gemma-4-31b showed no effect yet, consistent with the slow positive-sum effect in the long games (a post hoc reading; pooled: lower in 8 of 11 colony pairs, $p = 0.23$). A pre-registered punish test was inconclusive (fewer than 10 punishments in an arm); with the board 12 of 18 punishments hit the promise-breaker (chance 14%), all in one colony, against 1 of 4 without.

5. Discussion and Limitations

What fails. Our evidence points to a failure to represent and retain non-delivery: agents remember only what they write down; the promise gets written, the missing gift does not, and some later recall a broken promise as a gift. Stating non-delivery moves votes, and records that state it mitigate the failure; talk does not (the negative-sum colony complained most without voting differently). We cannot fully separate memory from inference, salience or the authority of the added sentence. That visible records sustain sanction is classic indirect reciprocity (Nowak & Sigmund 2005); what is new is where the failure sits when the agents are LLMs with self-written memory.

For multi-agent oversight. (1) Record commitments and deliveries explicitly and durably; even a private record did about half as much as a public one in our pilot. (2) Look at allocation, not only conversation: in every one of the 36 long games one member kept the shared store for 18 to 43 of 50 terms (one kept 51% of each store, four times a fair share), yet our crude count of mentions did not single it out (exploratory; with seed 0 everywhere, who rises may reflect name or seat). (3) Evaluate populations over long horizons with repeated measurement: within the 1000-day games the board's effect in the positive-sum economy took about 100 days to appear. (4) Do not assume persona effects on influence transfer between models: we found none that replicated.

Limitations

The 1000-day results rest on one model and 3 games per condition, all with seed 0, so the long-game board comparison has an effective sample nearer 3, and it was not pre-registered. The cross-model, private-ledger and punish results are 3-colony, 40-day pilots in the slowest economy. The promise-breaker is stylised (prompted, cheating daily, gifts blocked), and the 80-word self-rewritten notebook is a single memory design; longer or structured memory may behave differently. The designs differ in more than length (spoilage, reasons and poll frequency), so contrasts between them do not isolate duration. The promise-breaker writes its instruction in its notebook, so a judge model that reads notebooks detects a confession, not deception. Notebook and complaint coding are keyword-based, and the institution classifier is not yet validated. Placebos passed 0 of 96 tests where about 5 were expected, so the trait tests are conservative.

Future Work

Can a belief make itself true? The mirror image of the promise-breaker is a generous member whom some colonists have heard is selfish. Either the believers update as its gifts arrive, or the label comes true as the shunned target stops giving (behavioural confirmation; Merton 1948; Snyder et al. 1977;

between humans and AI, Pataranutaporn et al. 2023). The design is pre-registered (twin colonies without the rumor, the target's traits fixed, a private self-description probe), separates giving less only to believers (reciprocity) from giving less to everyone (a changed disposition), and runs are under way. Our mechanism predicts correction, because a gift that arrives leaves a trace; but a target that stays generous while believers keep misjudging it would be an equally important failure of collective oversight. We also plan independent colony set-ups for the record comparison and memory variants (longer notebooks, structured commitment records).

6. Conclusion

In a colony of near-identical LLM agents, persona text reliably changes how agents behave but not, across models, who gains influence. A member who breaks a promise every day keeps the colony's trust while the breach stays unstated, because self-written memory keeps the promise and drops the missing gift. A public record of gifts turns the colony against it over 1000 days, and a private one does about half of that. For populations of AI agents, making breaches explicit and watching who controls resources may matter more than the agents' own dispositions.

Code and Data

- **Code repository:** https://github.com/soham-padia/who-rises-in-the-colony (simulation colony_sim.py, every analysis script, pre-registrations in docs/).
- **Data:** Hugging Face dataset sohampadianeu/who-rises-in-the-colony-runs (every prompt and reply of every run); configs, game logs and agent tables are under runs/ in the repository.
- **Other artifacts:** a supplementary reference document with the full report, extended results, pre-registrations and the literature audit (docs/who_rises_in_the_colony_reference.pdf).

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Appendix

Supplementary document. The attached reference document (who_rises_in_the_colony_reference.pdf) holds the full 20-page report (all results, the trait and promise-breaker prompts, the economy table, the pre-registered catches and their outcomes), our research note, the 1000-day trend analysis and the literature review with its audit. Its later parts are working documents written during the hackathon; where they conflict with this report, this report is authoritative, and known superseded claims are listed in the research note.

Limitations and dual-use considerations. Limitations are discussed in Section 5. This work studies deception and influence among AI agents. The promise-breaker's instruction is a short, generic deception prompt, and the simulation touches no real system or person. The main dual-use risk is in the findings: they suggest how a deceiver avoids sanction (never let the breach become explicit; keep promising). We judge the defensive value larger, because the same results identify cheap countermeasures (persistent, shared records of commitments and deliveries; monitoring who controls resources). We release code and transcripts so others can test these defences, and report every pre-registered outcome, including nulls.

Reproducibility. 1000-day design: colony_sim.py --model gemma-4-31b-it --sizes 8 --games 3 --days 1000 --season-length 20 --spoil-above 15 --ask-why --economy positive|zero|negative [--public-ledger] [--byzantine]. 40-day design: --days 40 --season-length 10 [--private-ledger] [--

punish]. Pre-registrations: docs/preregistration.md (commit 8cf7fd0, amended in 20b2a18), docs/preregistration_punish.md, docs/preregistration_belief.md.

LLM Usage Statement

We used Claude (Anthropic), through Claude Code, extensively: to write and debug the simulation and analysis code, launch and monitor runs, compute statistics, check the literature and draft this report. Every number in the report is produced by scripts in the repository from the logged runs, and the team reviewed the claims against those outputs.